

# Quick Check

Lesson 6: Measuring Growth: Regions of Growth and Growth Curves of Maize H614 and Bean Rosecoco

*8 multiple-choice questions • choose A, B, C or D*

Sub-Strand 2.2: Growth and Development in Plants | Lesson 6

Fatuma's group marks a bean root with ink every 2 mm from the tip for 10 mm. After two days, the marks nearest the tip have spread furthest apart, while the marks further back are still close together. What does this show?

- A** The root grows in length mainly from the base near the seed
- B** The root grows in length equally along its whole length
- C** The root grows in length mainly in a region just behind the tip
- D** The root grows only in thickness and not in length

**A student says: 'Marks spreading near the root tip support the idea that new cells are made at the apical meristem and then elongate.' Which order of regions, starting from the very tip of a root, is correct?**

- A**    Region of elongation, region of cell division, region of differentiation
- B**    Region of differentiation, region of elongation, region of cell division
- C**    Region of cell division, region of differentiation, region of elongation
- D**    Region of cell division, region of elongation, region of differentiation

The mean shoot height of maize H614 was 62 cm at Day 28 and 98 cm at Day 35. What was the increase in height between Day 28 and Day 35?

A 160 cm

B 36 cm

C 27 cm

D 32 cm

The mean height of bean Rosecoco was 4, 9, 18, 30, 40 and 44 cm at Days 7, 14, 21, 28, 35 and 42. Between which two measurements was the increase in height greatest?

**A** Day 14 and Day 21

**B** Day 21 and Day 28

**C** Day 35 and Day 42

**D** Day 7 and Day 14

**Maize H614 increased by only 9 cm between Day 7 and Day 14 (3 cm to 12 cm), but by 36 cm between Day 28 and Day 35. Which phase of the growth curve does the first period show?**

- A** Stationary phase, because growth has stopped
- B** Lag phase, because growth is slow at the start
- C** Death phase, because the plant is weak
- D** Log phase, because growth is fastest

**Bean Rosecoco was 40 cm tall at Day 35 and 44 cm at Day 42, so its curve is flattening. What does the flattening show?**

- A** The bean plant has died
- B** The bean plant is in the log phase, where growth is fastest
- C** The bean plant is starting its lag phase
- D** The bean plant is in the stationary phase, where growth slows or stops

At Day 7, bean Rosecoco was 4 cm tall and maize H614 was 3 cm. At Day 42, maize was 125 cm and bean was 44 cm. A student says: 'Maize must have grown faster than bean all the time.' Which response is correct?

- A** Wrong, because bean was taller than maize at Day 7, so maize did not grow faster all the time
- B** Correct, because maize is taller at Day 42
- C** Wrong, because bean grew faster than maize between Day 28 and Day 42
- D** Correct, because maize roots are longer than bean roots

**A learner draws a graph of shoot height (y-axis) against time in days (x-axis) for maize H614 over 42 days. Which description matches the overall shape of a normal growth curve?**

- A** A curve that rises steeply at first and then falls back to zero
- B** A horizontal line
- C** A straight line sloping upwards
- D** An S-shaped (sigmoid) curve with lag, log and stationary phases

# All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).